TANIA RAMZI AJAB

Chouf, Lebanon | +961 81 078831 | Tania.gms2023@gmail.com | LinkedIn: tania-ajab

PhD Applicant | Medical AI Researcher | Computer Vision & Vision–Language Models

Professional Profile

PhD-oriented M.Sc. graduate in Computer & Communication Engineering with a strong research focus on medical artificial intelligence, vision–language models, computer vision, prompt learning, and trustworthy AI evaluation. My M.Sc. thesis investigates the failure modes of medical vision–language models under in-domain and out-of-domain settings, with emphasis on domain shift, calibration, prompt sensitivity, and prompt learning degradation in chest X-ray classification. I have hands-on experience in Python, PyTorch, machine learning pipelines, experimental design, statistical evaluation, scientific writing, and applied AI/software development. I am seeking a PhD opportunity where I can contribute to research in medical AI, multimodal learning, reliable AI systems, computer vision, healthcare technologies, or intelligent connected systems.

Research Interests

Medical Artificial Intelligence | Vision–Language Models | Multimodal Learning | Computer Vision | Trustworthy AI | Prompt Learning | Domain Shift | Model Calibration | Healthcare AI Safety | Biomedical Signal Processing | Brain–Computer Interfaces

Education

Rafik Hariri University — **Mechref, Lebanon**
Master of Science in Computer & Communication Engineering — *May 2026*

- Thesis: *Failure Analysis of Medical Vision–Language Models: Zero-Shot Performance, Prompt Sensitivity, and Prompt Learning Degradation in BiomedCLIP and MedCLIP.*
- Research focus: medical AI, vision–language models, CLIP-based models, prompt learning, calibration, domain shift, and chest X-ray pathology classification.
- CGPA: 90/100.

Rafik Hariri University — **Mechref, Lebanon**
Bachelor of Engineering in Computer & Communications Engineering — *May 2025*

- Dean's Honor List, 2023–2025.

Research Experience

M.Sc. Researcher – Medical Vision–Language Models — **2025 – 2026**
Rafik Hariri University — *Mechref, Lebanon*

- Thesis GitHub Repository: Medical VLM Failure Analysis
- Designed and executed a complete evaluation pipeline for medical vision–language models using Python, PyTorch, OpenCLIP, MedCLIP, NumPy, pandas, and scikit-learn.
- Evaluated BiomedCLIP, MedCLIP, MedKLIP, and CLIP-based baselines on medical image classification and biomedical figure-caption matching tasks.
- Compared in-domain and out-of-domain performance across PMC figure-caption data, CheXpert, and NIH ChestX-ray14.
- Analyzed five chest X-ray pathologies: Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion.
- Studied AUROC, AUPRC, F1 score, sensitivity at fixed specificity, Expected Calibration Error, threshold behavior, and bootstrap confidence intervals.
- Investigated prompt sensitivity, prompt ensembling, and prompt learning methods including CoOp, CoCoOp, VPT, and MaPLe.
- Produced publication-ready result tables, visualizations, statistical summaries, and a failure taxonomy for thesis writing and research reporting.

Publications & Conference Output

Machine Learning Based EEG Intent Classification for Smart Wheelchair Control — **Accepted, BIOSMART 2026**

- Developed a machine-learning framework for EEG-based brain–computer interface control of a smart wheelchair using a consumer-grade single-channel headset.

- Collected EEG data from 70+ volunteers across five imagined wheelchair commands: Forward, Backward, Left, Right, and Stop.
- Built preprocessing, segmentation, feature extraction, PCA, normalization, and XGBoost classification pipeline.
- Achieved 95.49% offline classification accuracy and connected the trained classifier to an Arduino-based control prototype.

Manuscripts in Preparation – M.Sc. Thesis Research

- Preparing two conference papers and one journal article based on my M.Sc. thesis on medical vision–language models, prompt sensitivity, domain shift, calibration, and failure-mode analysis.

Technical Skills

AI & Machine Learning	Deep Learning, Computer Vision, Vision–Language Models, Prompt Learning, Medical Image Analysis, YOLO, OpenCV, TensorFlow, PyTorch, scikit-learn, NLP fundamentals
Generative AI & RAG	Retrieval-Augmented Generation, prompt engineering, medical text processing, AI safety guardrails, citation-supported responses
Programming	Python, C++, C#, MATLAB, Verilog, SQL
Data & Evaluation	NumPy, pandas, Matplotlib, AUROC, AUPRC, F1 Score, calibration metrics, threshold analysis, bootstrap confidence intervals
Web Development	HTML5, CSS3, JavaScript, jQuery, Bootstrap, PHP, Laravel, JSON, API Development, Secure API Integration
Mobile Development	Kotlin, Android application development, Firebase integration, local storage, mobile monitoring interfaces
Databases	MySQL, SQLite, Firebase
Systems & IoT	Arduino, NodeMCU, Wemos, Raspberry Pi, embedded systems, sensors, motor control, IoT monitoring systems
Networking & Cybersecurity	TCP/IP, UDP, DNS, DHCP, HTTP/HTTPS, VPN, Wireshark, Cisco Packet Tracer, Kali Linux, Nmap, Burp Suite, data encryption
Tools	Git, Linux, LaTeX, technical documentation, project reporting

Selected Projects

AI Medical Report Explainer with Retrieval-Augmented Generation *2026 – Present*

- Developing a healthcare AI web application that helps users understand medical lab reports, such as blood tests and CBC reports, in simple patient-friendly language.
- Built functionality to upload medical reports and extract key values including test names, results, units, and reference ranges.
- Uses Natural Language Processing to identify medical values and classify results as low, normal, or high based on the report's reference ranges.
- Integrates Retrieval-Augmented Generation to search trusted medical sources and generate clear explanations supported by citations.
- Includes safety guardrails to prevent diagnosis, medication advice, or unsafe medical claims.
- Combines AI, NLP, RAG, healthcare AI safety, backend development, frontend development, database management, and deployment.

Mind-Controlled Smart Wheelchair / EEG-Based BCI Control *2024 – 2025*

- Built an assistive mobility prototype combining EEG intent classification, voice control, joystick control, obstacle detection, mobile monitoring, and Arduino-based actuation.
- Designed a low-cost EEG data pipeline for five imagined movement commands and integrated the trained classifier with wheelchair command mapping.
- Implemented mobile application features for monitoring SPO2, heart rate, emergency calls, and local data storage.
- Integrated real-time obstacle detection using ultrasonic sensors and a live-feed reversing camera.
- Technologies: Python, Kotlin, SQLite, Firebase, HTML, CSS, JavaScript, scikit-learn, Vosk, pyttsx3, Arduino.
- Demo: https://youtu.be/jfvDJUJ7VSY

Pothole Detection System *2024*

- Built a real-time pothole detection pipeline using YOLOv4 and computer vision techniques.
- Focused on detection accuracy, speed, and robustness under real-world road and lighting conditions.

Smart Agriculture Monitoring System *2024*

- Designed a secure IoT-based monitoring architecture with sensor integration, network design, and data analysis components.
- Supported automated environmental monitoring for agriculture-related decision-making.

Hospital Management System *2024*

- Developed a C# and MySQL-based hospital management system with database-driven workflows.
- Implemented structured modules for data storage, user interaction, and administrative process support.

Coffee & Book Shop Website *2024*

- Built a full-stack Laravel-based e-commerce platform.
- Developed front-end and back-end components with database integration and basic web security practices.

Automatic Pill Dispenser *2023*

- Designed an embedded system using Arduino, infrared sensors, LCD, and speaker components.

Traffic Management System *2024*

- Developed a Python-based system for dynamic traffic signal scheduling.

Professional & Teaching Experience

AI Chatbot Development Intern **Jun 2025 – Aug 2025**
Jaber Consulting *Lebanon*

- Developed an Arabic-language training center enrollment chatbot that helps users view course information, register for programs, and receive confirmation emails.
- Built both console-based and Telegram bot versions using Python and the Telegram Bot API.
- Integrated Gemini Flash 1.5 for AI-powered Arabic interaction and Gmail API for automated confirmation emails.
- Designed modular project files including main.py, config.py, bot$_{l}ogic.py, courses.json, gmail_{u}tils.py, andtelegrambot.py.$
- Implemented secure configuration handling using environment variables and strengthened understanding of OAuth 2.0 and Google Cloud APIs.
- Added user session handling, input validation, debugging improvements, UX polish, and final documentation for deployment readiness.

Freelance AI & Software Project Developer **Mar 2024 – Present**
Freelance *Lebanon*

- Designed and implemented AI-driven applications using machine learning, deep learning, computer vision, and image-processing techniques.
- Developed real-time detection systems using YOLO, OpenCV, and custom preprocessing pipelines.
- Built end-to-end workflows for data preprocessing, model training, evaluation, deployment, and documentation.
- Assisted students and clients with AI, MATLAB, Python, automation, software, and prototype-based academic projects.

Teaching Assistant / Lab Instructor **Present**
Rafik Hariri University *Mechref, Lebanon*

- Support networking physical laboratory sessions and assist students with lab implementation and troubleshooting.
- Help students understand networking concepts, practical configurations, and exam preparation topics.

Instructor / Teacher – Part Time **Present**
Ajial El Ghad College *Lebanon*

- Teach foundational courses in computer and communication engineering.
- Support students through guided problem-solving, technical explanations, exercises, and academic guidance.

Freelance Networking & Cybersecurity Consultant **Mar 2024 – Present**
Freelance *Lebanon*

- Assisted students with networking protocols, cybersecurity basics, penetration testing concepts, and lab-style practice.
- Worked with tools and topics including TCP/IP, VPN, DNS, Wireshark, Kali Linux, Nmap, and secure communication concepts.

Web Developer – Volunteer **Jan 2023 – Apr 2023**
Ajial El Ghad College *Lebanon*

- Designed and developed a front-end website using HTML, CSS, JavaScript, jQuery, and JSON.

- Improved website presentation, usability, and structure as part of a real implementation-ready project.

Workshop Organizer / Presenter **Jan 2022 – May 2025**
Rafik Hariri University *Lebanon*

- Presented workshops on Computer and Communication Engineering topics.
- Prepared learning materials and demonstrations for undergraduate audiences, including technical system examples.

Achievements

- Dean's Honor List at Rafik Hariri University, 2023–2025.
- Contributed to the "Smart Irrigate" proposal for the Microchip International Contest as part of a Rafik Hariri University team; the project was recognized among the top 20 globally and awarded a free Microchip AVR IoT chip.
- Past member in the Semicolon CTF Cybersecurity Competition, engaging in penetration testing, network security, and ethical hacking challenges.

Certifications

- CCNA 1–4 – Cisco Networking Academy *Sep 2023 – Apr 2024*
- Essential Computer Science Certificate *Sep 2021*
- Information Technology Basics Certificate *Jul 2024 – Aug 2024*
- Network Essentials Certificate *Sep 2024 – Nov 2024*
- Microsoft: Describe Cloud Computing *Dec 2024*
- Microsoft: Fundamental AI Concepts *Jan 2025*
- Microsoft: Core Architectural Components of Azure *Jan 2025*
- LinkedIn: How to Land a Job Interview *Dec 2024*
- LinkedIn: Become a Full Stack Developer *In Progress*
- Coursera: Fundamentals of Machine Learning and Artificial Intelligence *In Progress*

Soft Skills

Problem-solving | Critical thinking | Scientific communication | Technical writing | Teamwork | Teaching | Time management | Adaptability | Attention to detail

Languages

Arabic: Native **English:** Fluent **French:** Beginner

References

Available upon request.